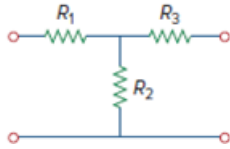


Problem

(a) For the T network in Fig. 19.106, show that the h parameters are:

$$\begin{aligned} h_{11} &= R_1 + \frac{R_2 R_3}{R_1 + R_3}, & h_{12} &= \frac{R_2}{R_2 + R_3} \\ h_{21} &= -\frac{R_2}{R_2 + R_3}, & h_{22} &= \frac{1}{R_2 + R_3} \end{aligned}$$

Figure 19.106

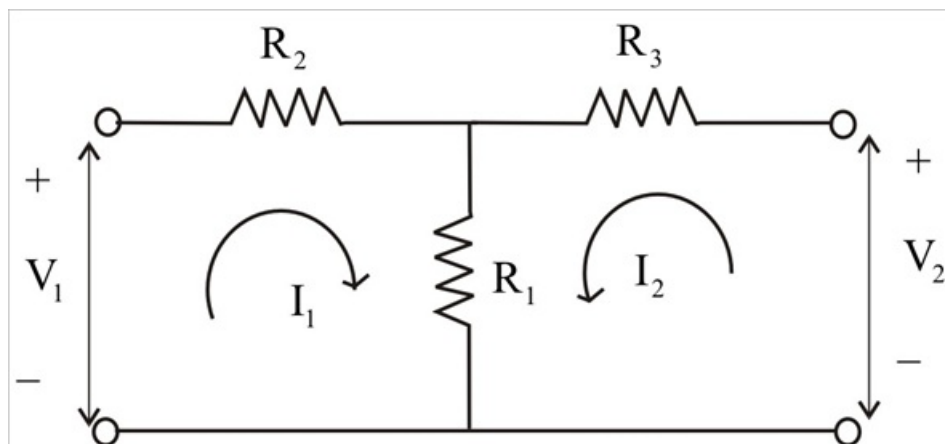


(b) For the same network, show that the transmission parameters are:

$$\begin{aligned} A &= 1 + \frac{R_1}{R_2}, & B &= R_3 + \frac{R_1}{R_2}(R_2 + R_3) \\ C &= \frac{1}{R_2}, & D &= 1 + \frac{R_3}{R_2} \end{aligned}$$

Step-by-step solution

Step 1 of 4



Step 2 of 4

Using mesh analysis we have

$$\begin{bmatrix} (R_1 + R_2) & R_2 \\ R_2 & (R_2 + R_3) \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}$$

$$\text{so, } [Z] = \begin{bmatrix} Z_{11} & Z_{12} \\ Z_{21} & Z_{22} \end{bmatrix} = \begin{bmatrix} (R_1 + R_2) & R_2 \\ R_2 & (R_2 + R_3) \end{bmatrix}$$

$$\text{so, } \Delta Z = (R_1 + R_2)(R_2 + R_3) - R_2^2 = R_1 R_2 + R_1 R_3 + R_2 R_3$$

Now

Step 3 of 4

$$\begin{aligned}
 \text{(a)} \quad [h] &= \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} = \frac{1}{Z_{22}} \begin{bmatrix} \Delta Z & Z_{12} \\ -Z_{21} & 1 \end{bmatrix} = \frac{1}{R_2 + R_3} \begin{bmatrix} (R_1 R_2 + R_1 R_3 + R_2 R_3) & R_2 \\ -R_2 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} \frac{R_1 (R_2 + R_3) R_2 R_3}{(R_2 + R_3)} & \frac{R_2}{(R_2 + R_3)} \\ \frac{-R_2}{(R_2 + R_3)} & \frac{1}{(R_2 + R_3)} \end{bmatrix} \\
 \text{or, } h_{11} &= R_1 \frac{R_2 R_3}{(R_2 + R_3)} \quad h_{12} = \frac{R_1}{R_2 + R_3} \quad h_{21} = -\frac{R_2}{R_2 + R_3} \quad h_{22} = \frac{1}{R_2 + R_3}
 \end{aligned}$$

Step 4 of 4

$$\begin{aligned}
 \text{(b)} \quad [T] &= \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \frac{1}{Z_{21}} \begin{bmatrix} Z_{11} & \Delta Z \\ 1 & Z_{22} \end{bmatrix} = \frac{1}{R_2} \begin{bmatrix} (R_1 + R_2) & (R_1 R_2 + R_1 R_3 + R_2 R_3) \\ 1 & (R_2 + R_3) \end{bmatrix} \\
 \Rightarrow A &= \frac{R_1 + R_2}{R_2} = 1 + \frac{R_1}{R_2} \\
 B &= \frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_2} = R_3 + \frac{R_1}{R_2} (R_2 + R_3) \\
 C &= \frac{1}{R_2} \\
 D &= \frac{R_2 + R_3}{R_2} = 1 + \frac{R_1}{R_2}
 \end{aligned}$$